Mehmet Aygun

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Education

- **University of Edinburgh** — Edinburgh, UK
  PhD in Computer Science (Computer Vision/ML) — *Sep 2021 – Jun 2025*
- **Technical University of Munich** — Munich, Germany
  MSc in Computer Science — *Oct 2018 – Dec 2020*
- **Istanbul Technical University** — Istanbul, Turkey
  BSc in Computer Science — *Sep 2013 – Jun 2017*

Experience

- **Genbio** — Paris, France
  AI Research Scientist — *Sep 2025 – Present*
  - Architecting and training **billion-parameter generative foundation models** for virtual cell simulation using Flow Matching and Self-Supervised Learning on massive biological datasets.
  - Spearheading the optimization of distributed training pipelines (FSDP/Kubeflow) to enable scalable 3D/4D biological representation learning.
- **Meta** — California, USA
  AI Research Scientist Intern — *Aug 2023 – Feb 2024*
  - Engineered a novel integration of **Self-Supervised Learning (SSL) with 3D Generative Models**, significantly improving reconstruction fidelity and zero-shot recognition.
  - Prototyped scalable training strategies for large-scale vision systems, optimizing the trade-off between representation quality and generative capability.
- **Bunsar** — Istanbul, Turkey
  Machine Learning Engineer — *Feb 2017 – Sep 2018*
  - Deployed large-scale production CV systems for fashion: image retrieval, object detection, and multi-object tracking.
  - Optimized inference latency and model quantization for real-time performance in high-traffic production environments.

Selected Publications

- S. Kapse, **M. Aygun**, et al. *GenBio-PathFM: A Foundation Model for Histopathology*. Under Review, 2026.
- D. Duolikun, **M. Aygun**, et al. *DepthCues: Evaluating Monocular Depth Perception in Large Vision Models*. CVPR 2025.
- **M. Aygun**, O. Mac Aodha. *SAOR: Single-view Articulated Object Reconstruction*. CVPR 2024.
- **M. Aygun**, O. Mac Aodha. *Demystifying Unsupervised Semantic Correspondence Estimation*. ECCV 2022.
- **M. Aygun**, et al. *4D Panoptic LiDAR Segmentation*. CVPR 2021.

Technical Skills

- **Research:** Generative Modeling (Diffusion, Flow Matching), Self-Supervised Learning (MAE, DINO, JEPA), 3D Vision.
- **Engineering:** PyTorch (Distributed/Multi-GPU), C++, CUDA, Linux/Slurm/Kubeflow, Large-scale Data Pipelines.
- **Tools:** NumPy, SciPy, OpenCV, Git, Docker, WandB/TensorBoard.

Activities & Service

- **Organizer:** Fine-Grained Visual Categorization Workshop (CVPR 2025).
- **Reviewer:** TPAMI, CVPR, ICCV, ECCV, NeurIPS.
- **Teaching:** Graduate TA for Applied Machine Learning (UoE), 2021–2024.